



SEMESTER LONG INTERNSHIP

(AUTUMN)2026

FOSSEE eSim PCB Design Project Report

PROJECT TITLE:

Theremin Synth Voice (Op-Amp Oscillator)

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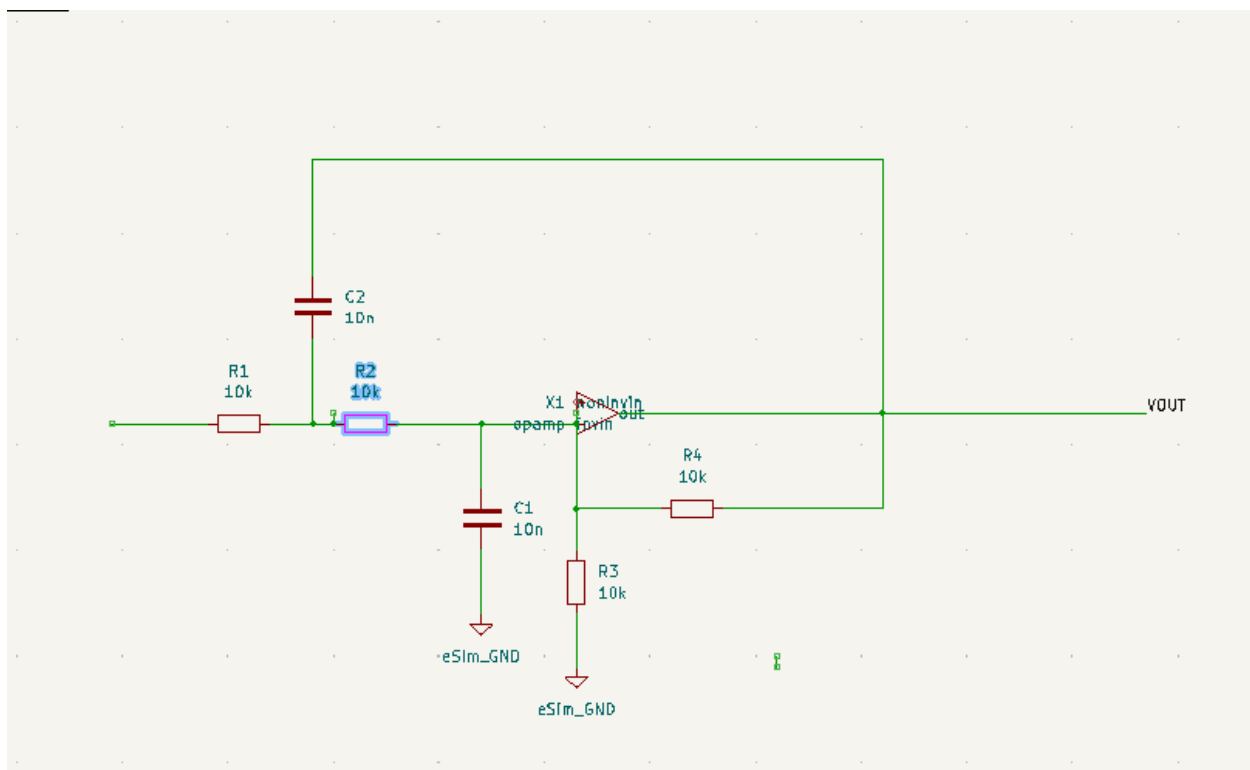
INSTITUTE: Vellore Institute of Technology (VIT) Bhopal University

BRANCH: ELECTRONICS AND COMMUNICATION ENGINEERING

YEAR: 3rd(6th semester)

1. Project Overview

This project involves the design and development of a custom Theremin Synth Voice (Op-Amp Oscillator). The primary objective was to implement a complete, industry-standard PCB design workflow using eSim (KiCad-based environment), taking the project from initial schematic capture to a fully routed, multi-layer physical layout. Special attention was given to signal integrity, proper footprint mapping, and minimizing parasitic inductance through optimal component placement and multi-layer routing strategies



2. Circuit Description & Footprint Assignment

The evaluation circuit is structured to support the core functionality and safe operation of the oscillator module. The circuit architecture consists of the following key components:

- **X1 (Op-Amp):** Configured with an 8-pin Dual In-line Package (Package_DIP: DIP-8_W7.62mm), serving as the central processing and interface unit.
- **C1, C2 (Capacitors):** 10nF through-hole capacitors (Capacitor_THT:CP_Radial_D5.0mm_P2.50mm) positioned specifically to filter high-frequency noise and set the oscillation time constant.
- **R1, R2, R3, R4 (Resistors):** 10k Ω axial resistors (Resistor_THT:R_Axial_DIN0204_L3.6mm_D1.6mm_P7.62mm_Horizontal) for current regulation and feedback loop control

Symbol : Footprint Assignments

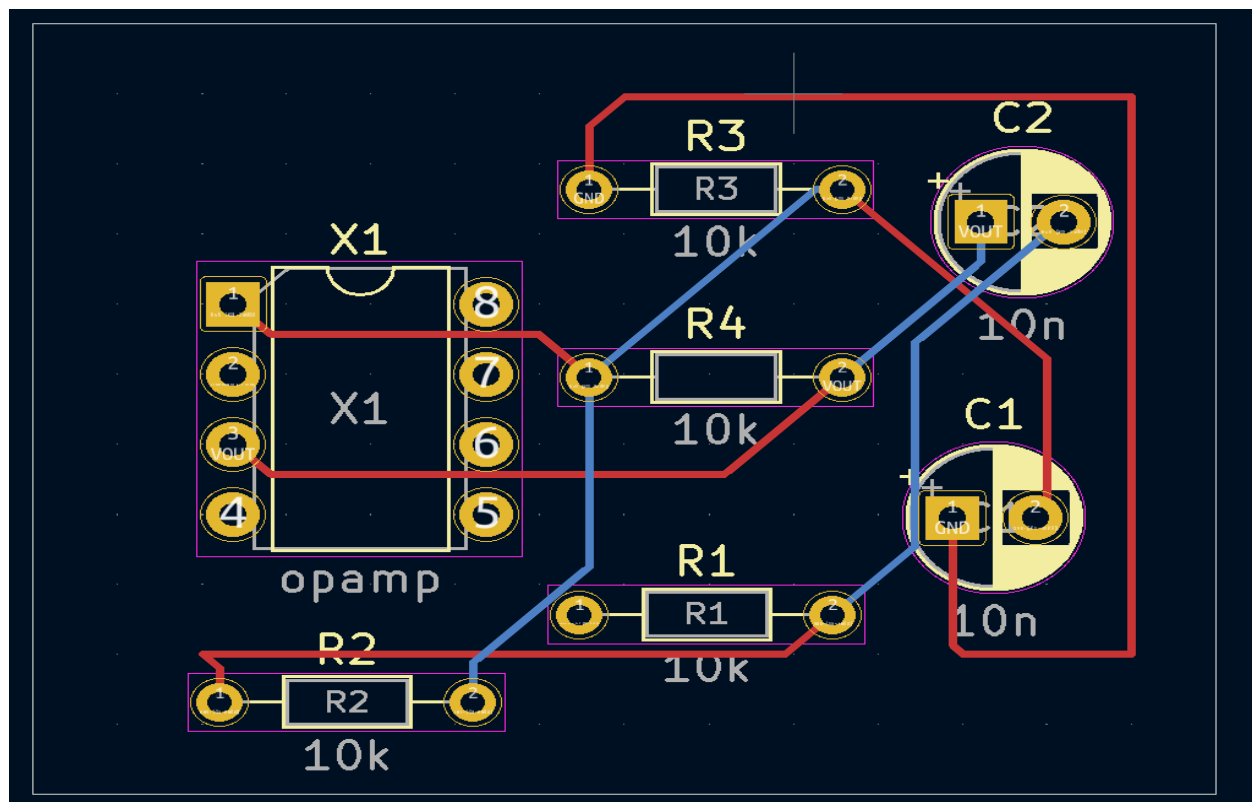
1	C1 -	10n : Capacitor_THT:CP_Radial_D5.0mm_P2.50mm
2	C2 -	10n : Capacitor_THT:CP_Radial_D5.0mm_P2.50mm
3	R1 -	10k : Resistor_THT:R_Axial_DIN0204_L3.6mm_D1.6mm_P7.62mm_Horizontal
4	R2 -	10k : Resistor_THT:R_Axial_DIN0204_L3.6mm_D1.6mm_P7.62mm_Horizontal
5	R3 -	10k : Resistor_THT:R_Axial_DIN0204_L3.6mm_D1.6mm_P7.62mm_Horizontal
6	R4 -	10k : Resistor_THT:R_Axial_DIN0204_L3.6mm_D1.6mm_P7.62mm_Horizontal
7	X1 -	opamp : Package_DIP:DIP-8_W7.62mm

3. Routing Methodology & Multi-Layer Design

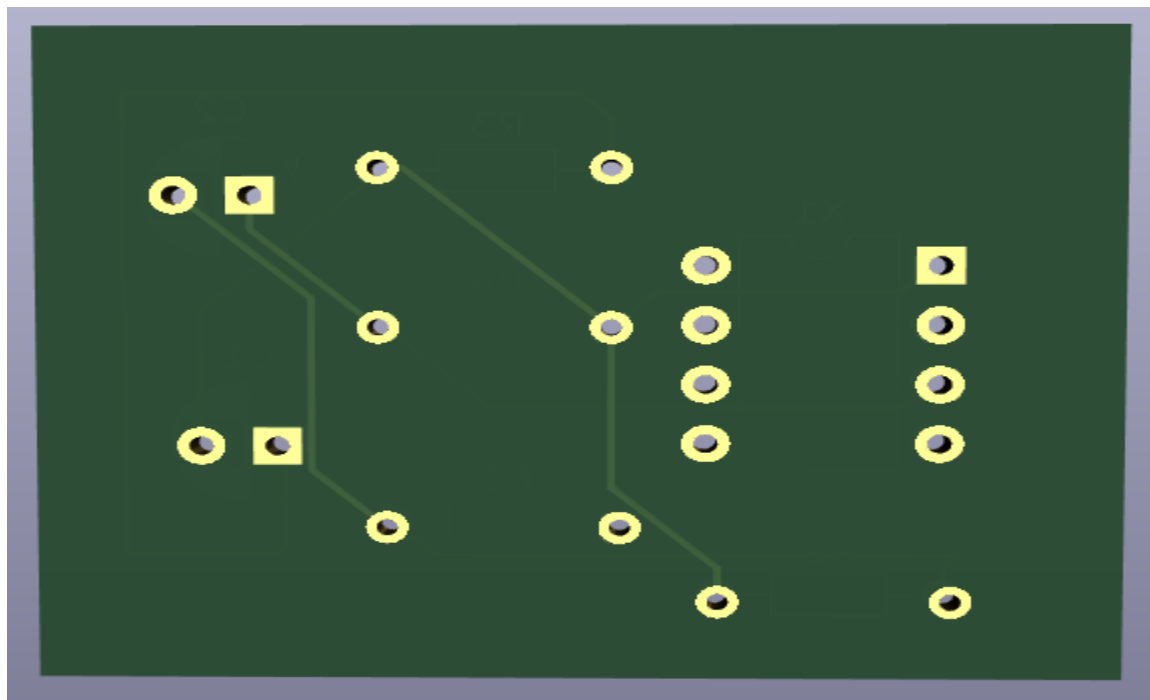
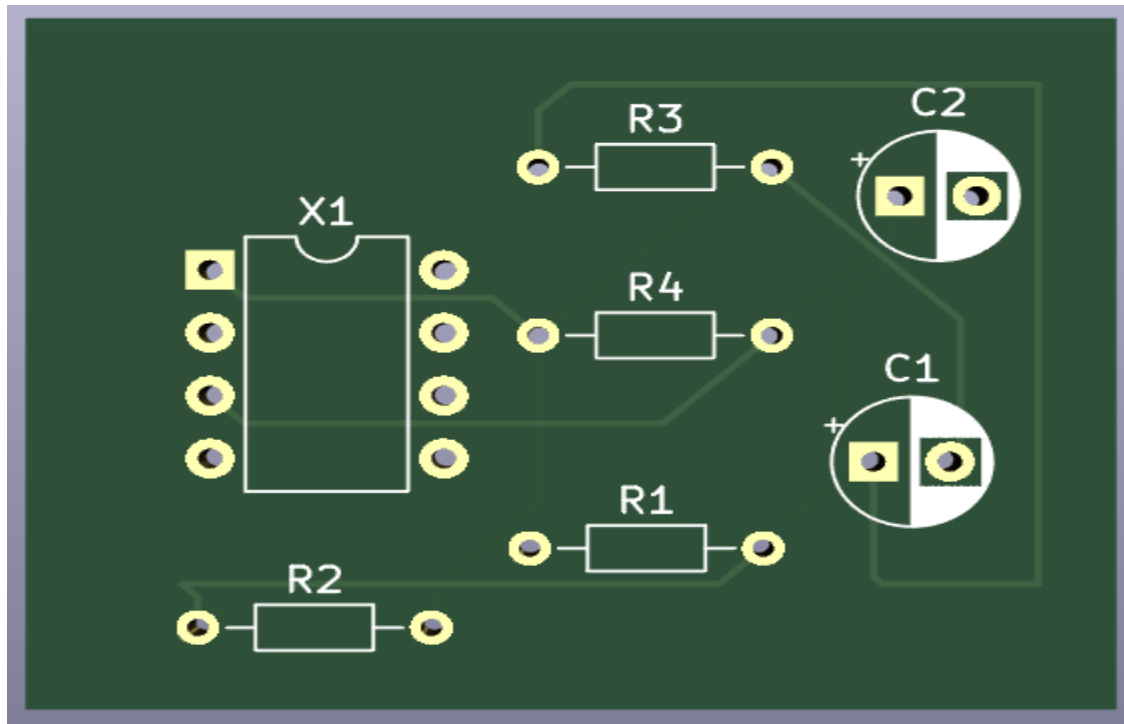
To achieve a clean layout without violating physical or electrical constraints, the board was implemented as a multi-layer PCB utilizing both top (F. Cu) and bottom (B. Cu) copper layers:

- **Component Placement:** The capacitors were placed in close proximity to the op-amp pins to reduce trace length and prevent parasitic inductance issues that could otherwise induce high-frequency signal noise.
- **Trace Routing & Crossovers:** By leveraging the bottom copper layer (B. Cu), blue routing tracks were passed safely underneath existing top-layer tracks (F. Cu). This eliminated overlapping trace errors and ensured clean, direct connectivity following the ratsnest guide lines.

2D PCB LAYOUT:

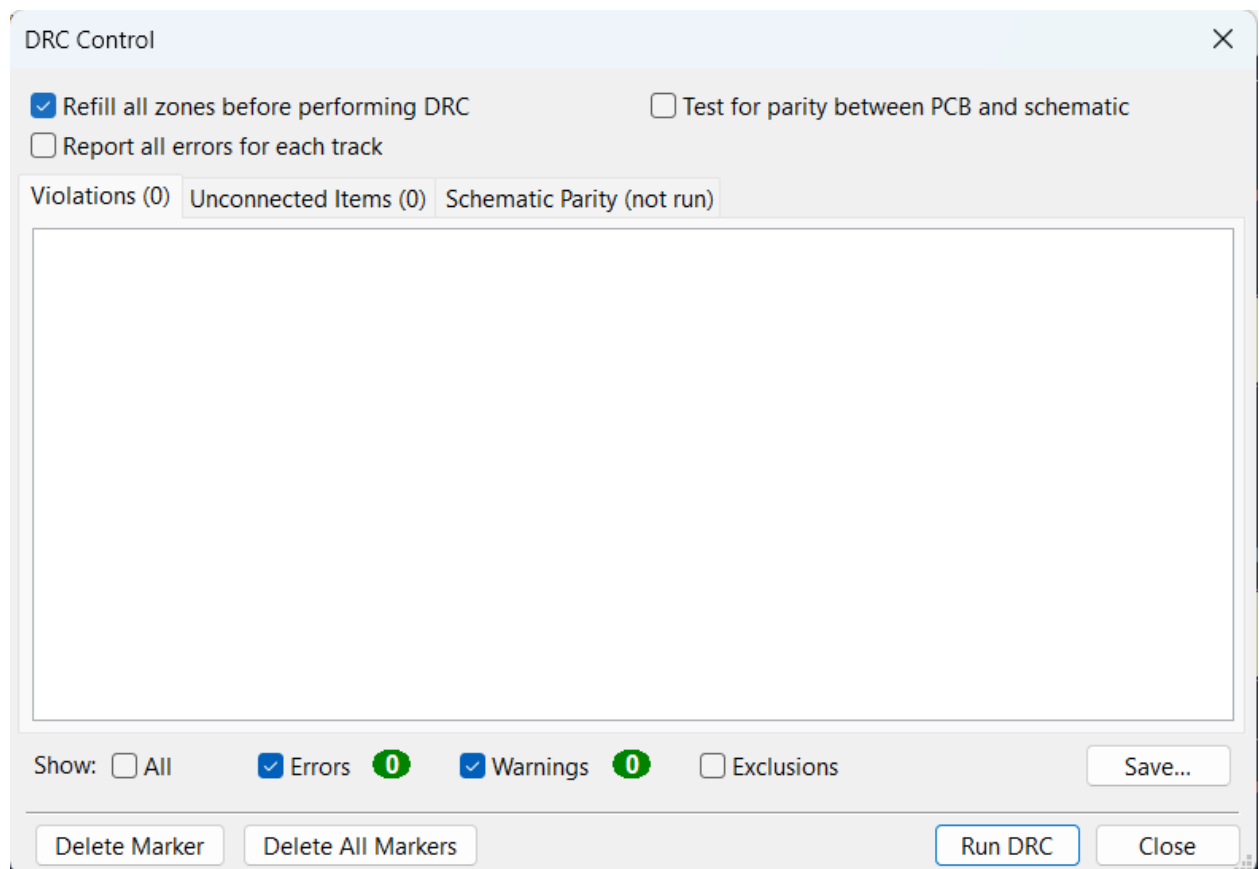


3D PCB LAYOUT:



4. Design Rule Check (DRC) Results:

- A comprehensive Design Rule Check (DRC) was executed within the eSim PCB editor to validate manufacturing compliance.
- All trace clearances, track widths, and pad-to-hole ratios were verified against standard manufacturing tolerances.
- Stray track stubs and dangling ends were systematically cleaned up.
- The final DRC execution returned 0 Errors and 0 Unconnected Items, confirming that the physical layout matches the logical schematic with absolute integrity.



5. Conclusion

The Theremin Synth Voice project has been successfully engineered, routed, and validated. Every mandatory workflow requirement—from eSchema capture and footprint assignment to multi-layer routing and DRC verification—was meticulously fulfilled using eSim. The resulting design is fully optimized, physically sound, and completely ready for manufacturing and FOSSEE project evaluation.